DuocUC

Ingeniería de soluciones con AI

Semestre 2026-2

Docente:

Francisco Macaya

Noticias: *Magnifica Humanitas, reacción del Papa sobre la AI.*



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Friday May 29, 2026

'Magnifica Humanitas': AI and the Pope

by BENEDIKT KOEHLER

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REPASO

¿Cuál es la diferencia entre la inteligencia artificial que uso en visual studio code o el chatGPT y el código de mi agente de la evaluación 2?

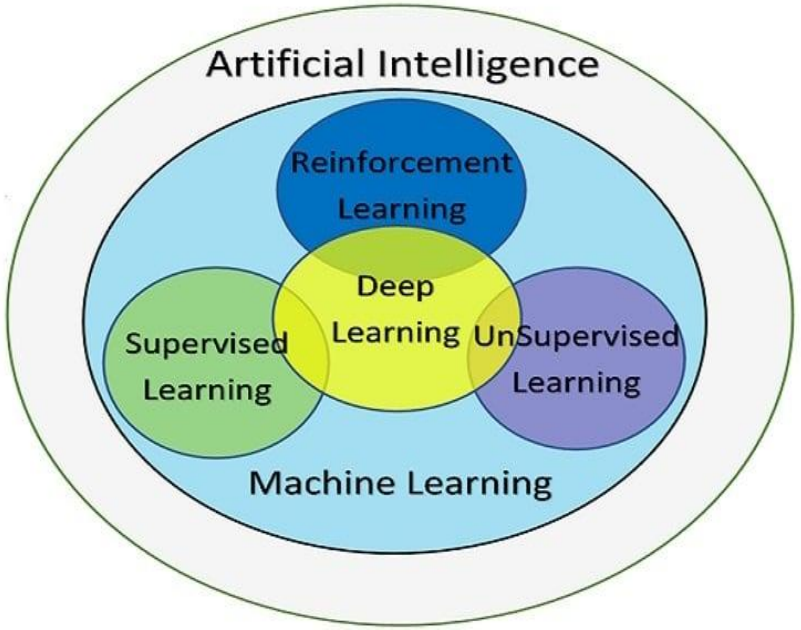
REPASO

¿Qué es un sistema multi-agente?

https://github.com/Foco22/INGENIERIA-DE-SOLUCIONES-CON-INTELIGENCIA-ARTIFICIAL_002D_OLS/tree/main/Experiencia%20de%20Aprendizaje%202%20-%20Desarrollo%20de%20Agente%20Inteligentes%20con%20AI/Clase%202.3

¿Cuál es la diferencia entre AI y ML?

REPASO

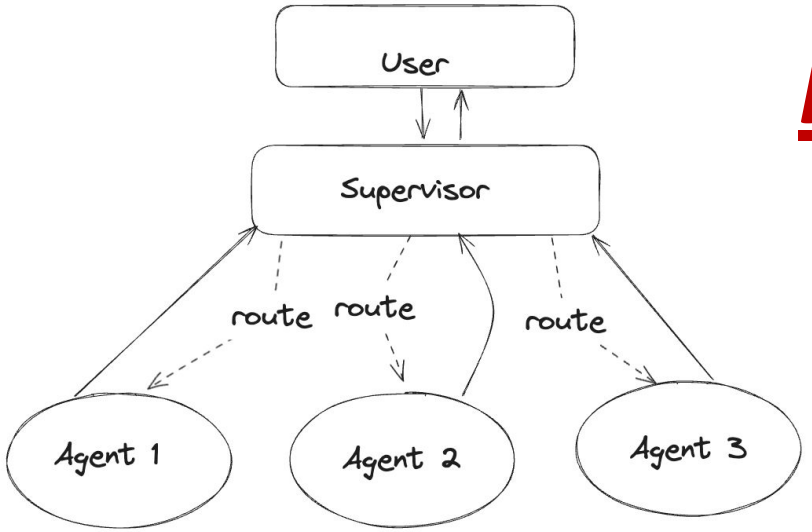


Artificial intelligence (AI) is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy

https://www.ibm.com/think/topics/artificial-intelligence

¿Cuál arquitectura es más escalable?

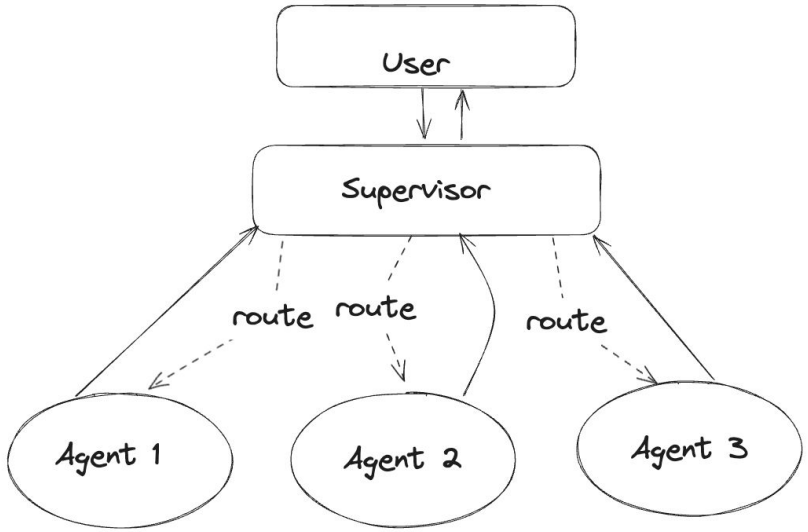
Plan A



REPASO

El agente supervisor decide a qué nodo dirigirse en función de las palabras proporcionadas por el usuario. Por ejemplo, si el usuario utiliza términos como "comprar", "compras", "comprarías" o similares, el supervisor selecciona el Agente A.

Plan B



El agente supervisor invoca a un LLM, que en base a un prompt de sistema, y una instrucción clara, decide a que agente debe llamar en cada momento.

REPASO

DuocUC

¿Cómo se escribe un endpoint (API) correctamente?

REST best practices (1/2)

- Use forward slash (/) to indicate hierarchical relationships
  - `http://stelios.io/api`
  - `http://stelios.io/api/users`
  - `http://stelios.io/api/users/{id}`
- Do not use trailing forward slash (/) in URIs
  - `http://stelios.io/api/users/` `/*Do not use this*/`
  - `http://stelios.io/api/users` `/*This is better*/`
- Use hyphens (-) to improve the readability of URIs, do not use underscores (_)
  - `http://stelios.io/api/users/mini_film/{id}` `/*Do not use this */`
  - `http://stelios.io/api/users/mini-film/{id}` `/*This is better*/`

REST best practices (2/2)

- Use lowercase letters in URIs
- Do not use file extensions
  - `http://stelios.io/api/users/user-data.json` `/* Do not use this */`
- Never use CRUD function names in URIs
  - `http://stelios.io/api/users/get-users` `/* Do not use this */`
  - `http://stelios.io/api/users` `/* This is better */`

Escalabilidad de un sistema: *Mas maquinas o mas potencia.*

Horizontal vs Vertical scaling

- Horizontal: Add more instances on a cluster
- Vertical: Increase the size of the virtualized resource (e.g., CPU, RAM, etc.)





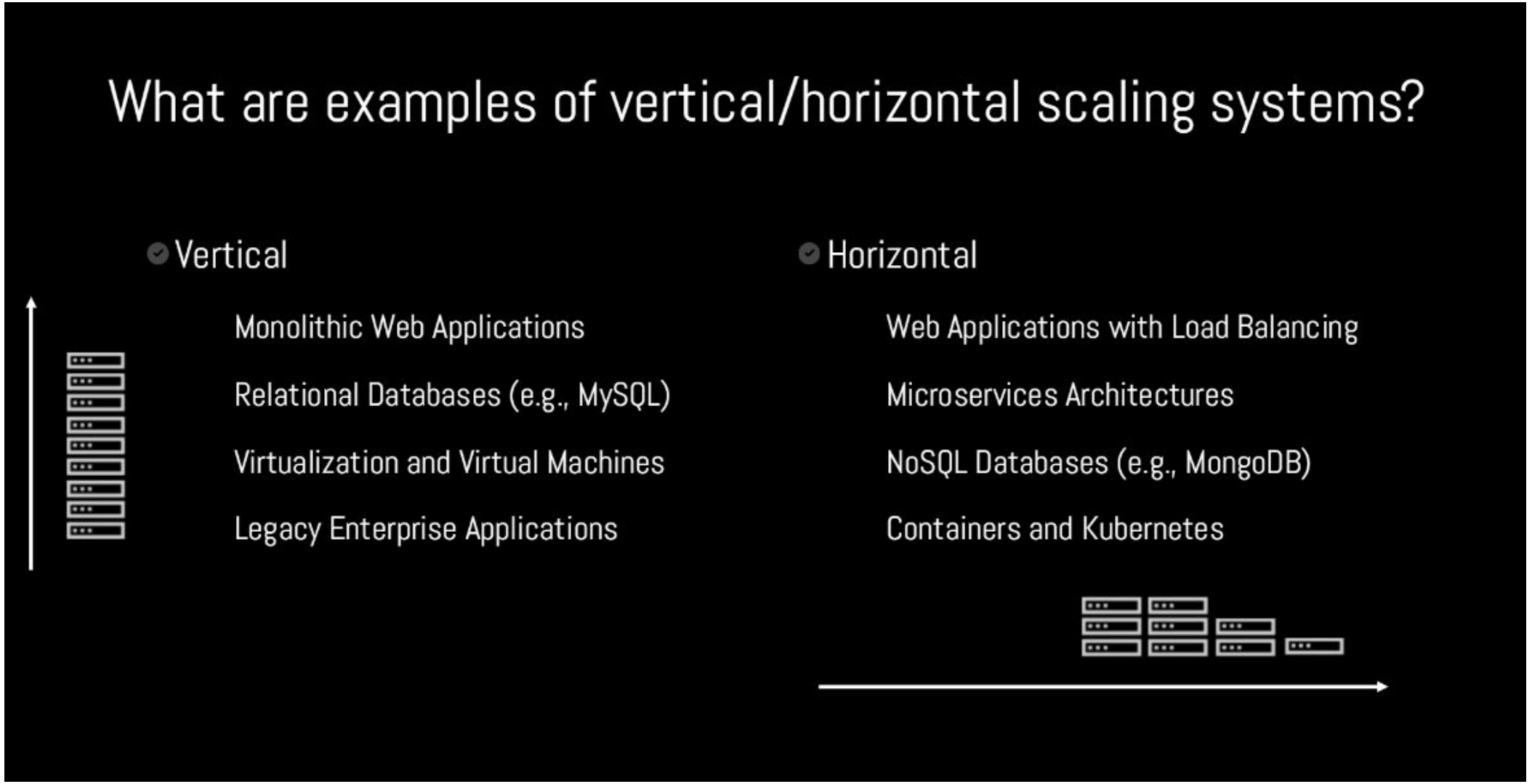
What is the problem with Vertical?

- Servers have limits

What is the problem with Horizontal?

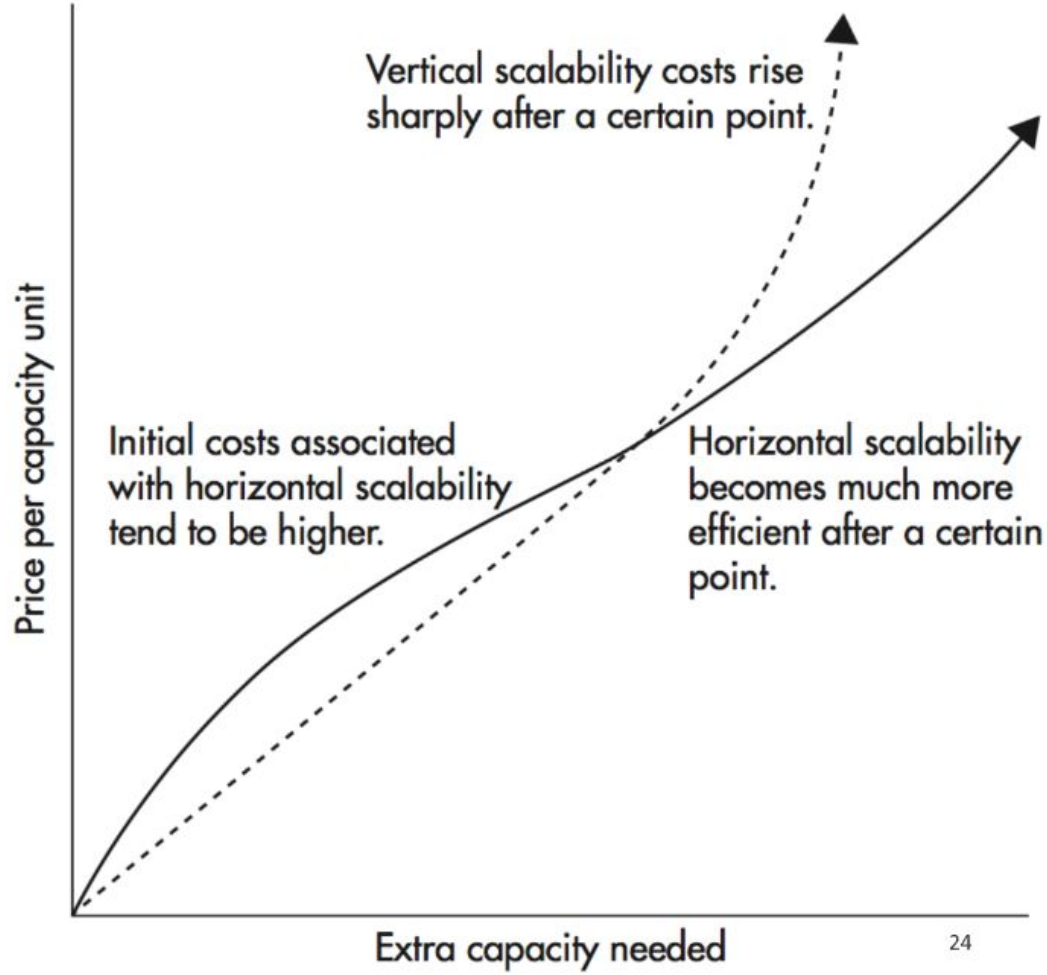
- It's a distributed system

Ejemplos: *Un start-up, con pocos usuarios, suele ser monolito.*



¿Cuánto moverse?: *Suele ser una decisión económica.*

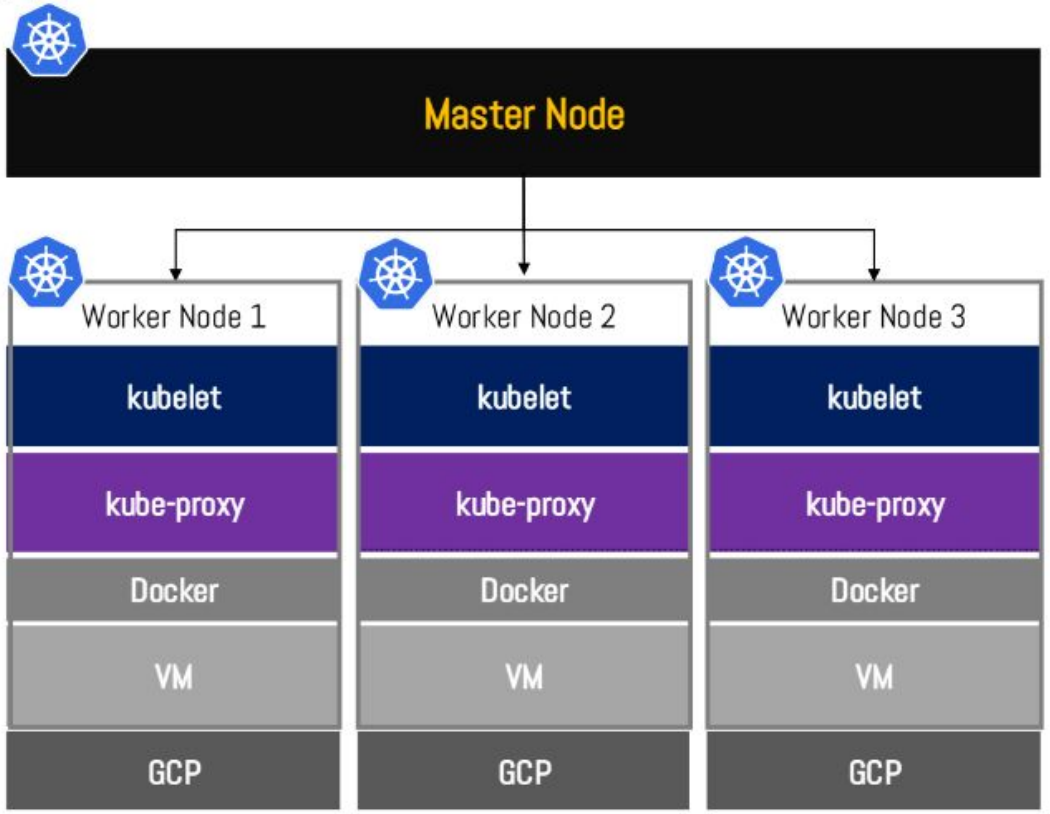
Vertical vs Horizontal cost



Kubernetes: *La solución escalable que soluciona cualquier tipo de problema de conexión entre VMs.*

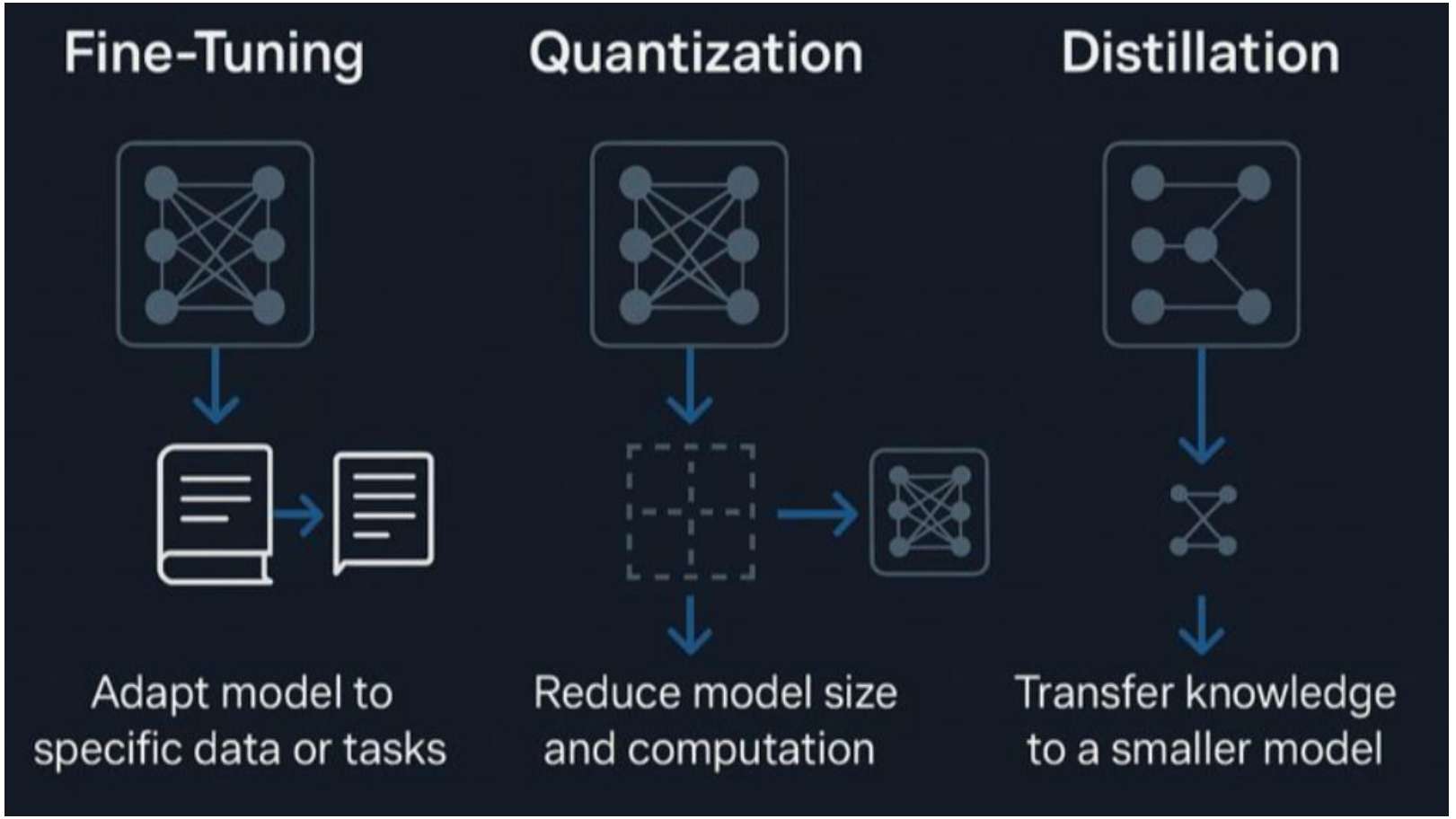
The Kubernetes system

- The master node(s) host the control plane aspects of the cluster
  - Kubernetes API server
  - Scheduler
  - Controller
  - Etcd
- Workers are the workforce, they run our containers!



Este ramo no es sobre Kubernetes. No es complicado configurarlo, ya que hoy en día, con GCP, es bastante fácil montar estos servicios, el problema es el costos, y que aplica solo a empresas que tienen un flujo de información alta, no a pequeñas empresas.

Optimización por modelamiento: *Hazlo más pequeño.*



Nota*: https://www.linkedin.com/pulse/fine-tuning-quantization-distillation-three-very-ways-alvin-kabwama-i47zc/.*

Optimización por inferencia de servicio: *GPU más eficiente.*

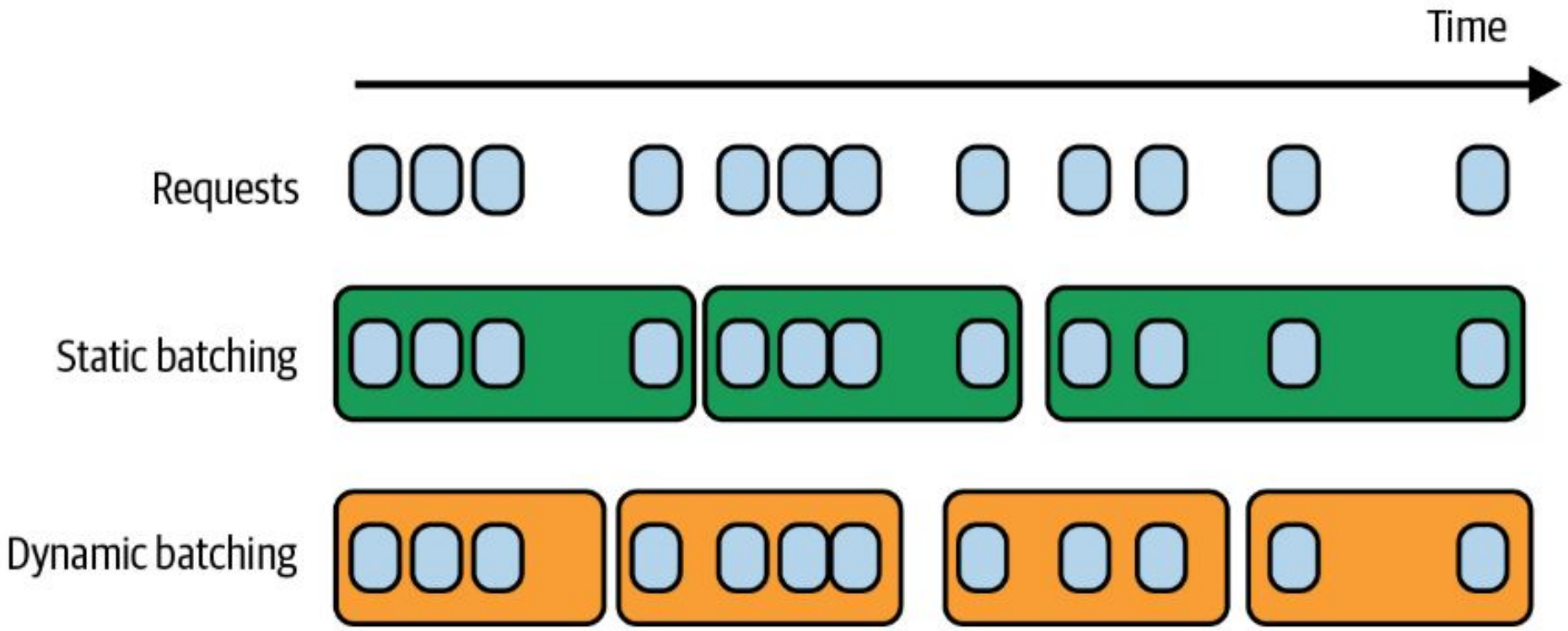


Figure 9-15. Dynamic batching keeps the latency manageable but might be less compute-efficient.

Requests esperan como en un Bus en ***Static batching****, y en* ***Dynamic batching****, esperan como en un bus, o hasta un cierto tiempo.*

Prompt caching: *No te repitas a ti mismo.*

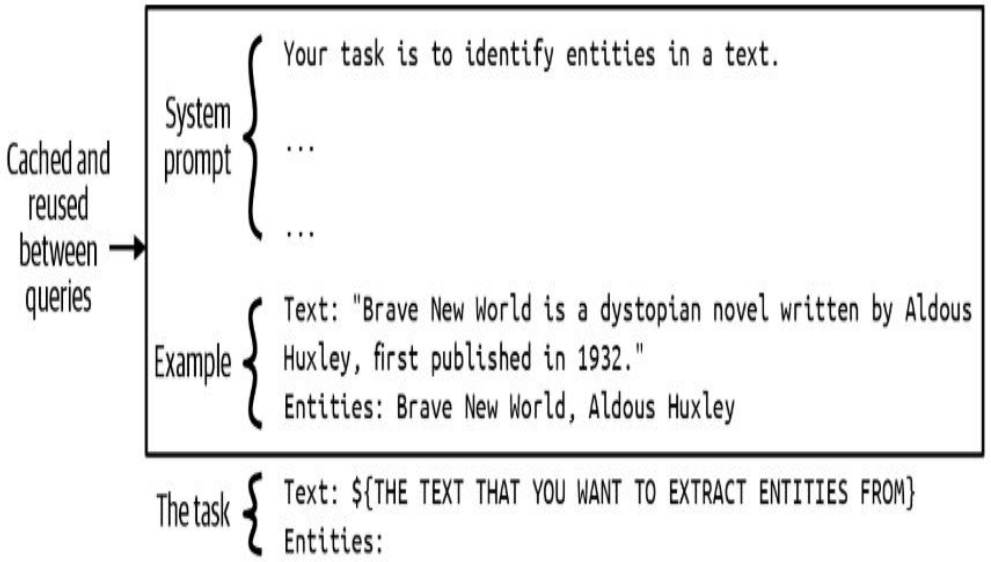


Figure 9-17. With a prompt cache, overlapping segments in different prompts can be cached and reused.

Table 9-3. Cost and latency reduced by prompt caching. Information from Anthropic (2024).

Use case	Latency w/o caching (time to first token)	Latency with caching (time to first token)	Cost reduction
Chat with a book (100,000-token cached prompt)	11.5 s	2.4 s (–79%)	–90%
Many-shot prompting (10,000-token prompt)	1.6 s	1.1 s (–31%)	–86%
Multi-turn conversation (10-turn convo with a long system prompt)	~10 s	~2.5 s (–75%)	–53%

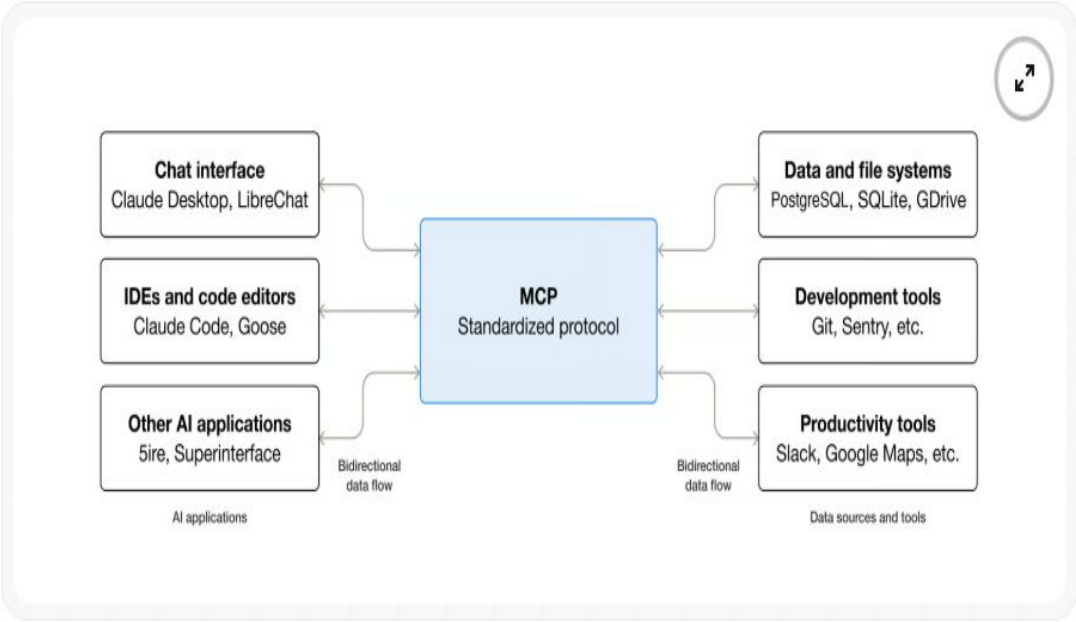
Prompt caching esta disponible por los proveedores de APIs de LLMs, y es una buena forma de optimizar costos.

MCP: *Model Context Protocol*

MCP (Model Context Protocol) is an open-source standard for connecting AI applications to external systems.

Using MCP, AI applications like Claude or ChatGPT can connect to data sources (e.g. local files, databases), tools (e.g. search engines, calculators) and workflows (e.g. specialized prompts)—enabling them to access key information and perform tasks.

Think of MCP like a USB-C port for AI applications. Just as USB-C provides a standardized way to connect electronic devices, MCP provides a standardized way to connect AI applications to external systems.



https://modelcontextprotocol.io/docs/2026-07-28/getting-started/intro

MCP: *¿Por qué importa y para qué sirve?*

What can MCP enable?

- Agents can access your Google Calendar and Notion, acting as a more personalized AI assistant.
- Claude Code can generate an entire web app using a Figma design.
- Enterprise chatbots can connect to multiple databases across an organization, empowering users to analyze data using chat.
- AI models can create 3D designs on Blender and print them out using a 3D printer.

Why does MCP matter?

Depending on where you sit in the ecosystem, MCP can have a range of benefits.

- **Developers**: MCP reduces development time and complexity when building, or integrating with, an AI application or agent.
- **AI applications or agents**: MCP gives them access to an ecosystem of data sources, tools and apps, which enhances their capabilities and improves the end-user experience.
- **End-users**: MCP results in more capable AI applications or agents that can access user data and take actions on the user's behalf when necessary.

https://modelcontextprotocol.io/docs/2026-07-28/getting-started/intro

Oxford Tech Week: *Agentes, Iot y personas.*



